
UNIVERSITY OF THE PHILIPPINES DILIMAN
College of Science
National Institute of Physics

Physics 180 THV-1
Problem Set 6-7

VERCIL JUAN
IV - BS Physics

September 4, 2026

Problem 6.1

Find the **second-order** and the **third-order** terms in the transition matrix element T_{fi} of Fermi's golden rule.

Solution

We start from the expansion of the state in terms of the eigenstates $|\phi_k\rangle$ of the unperturbed Hamiltonian,

$$|\psi(t)\rangle = \sum_k c_k(t) |\phi_k\rangle e^{-iE_k t}. \quad (1.1)$$

Substituting (1.1) into the time-dependent Schrödinger equation with interaction Hamiltonian $\hat{H}'$ gives

$$i \sum_k \frac{dc_k}{dt} |\phi_k\rangle e^{-iE_k t} = \sum_k \hat{H}' c_k(t) |\phi_k\rangle e^{-iE_k t}. \quad (1.2)$$

Taking the inner product with $\langle\phi_f|$ gives the differential equation for the coefficient $c_f(t)$ corresponding to transitions to the particular final state $|f\rangle = |\phi_f\rangle$. Under the first-order approximation used in the lecture, $c_i(t) \approx 1$ and $c_{k \neq i}(t) \approx 0$, giving

$$\frac{dc_f}{dt} = -i \langle f | \hat{H}' | i \rangle e^{i(E_f - E_i)t}. \quad (1.3)$$

The lecture defines the transition matrix element by

$$T_{fi} = \langle f | \hat{H}' | i \rangle. \quad (1.4)$$

To obtain the higher-order terms, we no longer restrict the sum in (1.2) to the initial state. We write the coefficients as an expansion in orders of the perturbing Hamiltonian,

$$c_f(t) = c_f^{(0)}(t) + c_f^{(1)}(t) + c_f^{(2)}(t) + c_f^{(3)}(t) + \dots. \quad (1.5)$$

The zeroth-order coefficients satisfy

$$c_i^{(0)}(t) = 1, \quad c_{k \neq i}^{(0)}(t) = 0. \quad (1.6)$$

For an intermediate state $|n\rangle$, the first-order coefficient follows from the same differential equation as (1.3),

$$\frac{dc_n^{(1)}}{dt} = -i \langle n | \hat{H}' | i \rangle e^{i(E_n - E_i)t}. \quad (1.7)$$

Integrating (1.7) from 0 to t gives

$$\begin{aligned} c_n^{(1)}(t) &= -i \langle n | \hat{H}' | i \rangle \int_0^t dt' e^{i(E_n - E_i)t'} \\ &= -\langle n | \hat{H}' | i \rangle \frac{e^{i(E_n - E_i)t} - 1}{E_n - E_i}. \end{aligned} \quad (1.8)$$

We now substitute (1.8) into the equation for $c_f(t)$. The second-order contribution is now therefore

$$\frac{dc_f^{(2)}}{dt} = -i \sum_n \langle f | \hat{H}' | n \rangle e^{i(E_f - E_n)t} c_n^{(1)}(t). \quad (1.9)$$

Using (1.8), we have

$$\frac{dc_f^{(2)}}{dt} = i \sum_n \frac{\langle f | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{E_n - E_i} \left[e^{i(E_f - E_i)t} - e^{i(E_f - E_n)t} \right]. \quad (1.10)$$

The first term in (1.10) has the same time dependence as the first-order result in (1.3). This is the term that contributes to the transition matrix element for the transition from $|i\rangle$ to $|f\rangle$. Thus, comparing with (1.3), the second-order transition matrix element is

$$T_{fi}^{(2)} = \sum_n \frac{\langle f | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{E_i - E_n} \quad (1.11)$$

Hence, to second order, the transition proceeds through an intermediate state $|n\rangle$,

$$|i\rangle \rightarrow |n\rangle \rightarrow |f\rangle. \quad (1.12)$$

For the third-order term, we repeat the same procedure. The second-order coefficient for an intermediate state $|m\rangle$ contains a contribution through another intermediate state $|n\rangle$,

$$c_m^{(2)}(t) \supset \sum_n \frac{\langle m | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{(E_i - E_m)(E_i - E_n)} e^{i(E_m - E_i)t}. \quad (1.13)$$

Substituting (1.13) into the coefficient equation gives the contribution to $c_f^{(3)}$ with the same time dependence as the first-order transition,

$$\frac{dc_f^{(3)}}{dt} \supset -i \sum_{m,n} \frac{\langle f | \hat{H}' | m \rangle \langle m | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{(E_i - E_m)(E_i - E_n)} e^{i(E_f - E_i)t}. \quad (1.14)$$

Comparing (1.14) with the form of the first-order differential equation in (1.3), the third-order transition matrix element is therefore

$$T_{fi}^{(3)} = \sum_{m,n} \frac{\langle f | \hat{H}' | m \rangle \langle m | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{(E_i - E_m)(E_i - E_n)} \quad (1.15)$$

Therefore, extending the transition matrix element defined in (1.4) to higher orders,

$$\begin{aligned} T_{fi} &= T_{fi}^{(1)} + T_{fi}^{(2)} + T_{fi}^{(3)} + \dots \\ &= \langle f | \hat{H}' | i \rangle + \sum_n \frac{\langle f | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{E_i - E_n} + \sum_{m,n} \frac{\langle f | \hat{H}' | m \rangle \langle m | \hat{H}' | n \rangle \langle n | \hat{H}' | i \rangle}{(E_i - E_m)(E_i - E_n)} + \dots \end{aligned} \quad (1.16)$$

Problem 7.1

The semi-major axis a and semi-minor axis b of a prolate spheroid with eccentricity x are related through

$$b = \sqrt{1 - x^2}a \quad (2.1)$$

If the volume and surface area of the prolate spheroid are

$$V_{\text{prolate}} = \frac{4}{3}\pi ab^2 \quad (2.2)$$

$$S_{\text{prolate}} = 2\pi b \left(b + \frac{a \sin^{-1}(x)}{x} \right) \quad (2.3)$$

and defining $\varepsilon = \frac{1}{3}x^2$, show that the equations in slide 36 (reproduced below) hold for small values of x .

$$a_2 A^{\frac{2}{3}} \rightarrow a_2 A^{\frac{2}{3}} \left(1 + \frac{2}{5}\varepsilon^2 \right) \quad (2.4)$$

$$a_3 \frac{Z^2}{A^{\frac{1}{3}}} \rightarrow a_3 \frac{Z^2}{A^{\frac{1}{3}}} \left(1 - \frac{1}{5}\varepsilon^2 \right) \quad (2.5)$$

Solution

We tackle the surface and coulomb term separately. A key observation here is that a and b must be adjusted so that the volume of a corresponding sphere remains constant. The SEMF equations are fundamentally based on the assumption of a spherical nucleus, and as such changes on that assumption means the terms scale respectively.

Let R be the radius of the sphere having the same volume as the spheroid. From (2.2), we have

$$\frac{4}{3}\pi ab^2 = \frac{4}{3}\pi R^3$$

Using (2.1), $b = a\sqrt{1 - x^2}$, we have

$$a^3(1 - x^2) = R^3$$

and hence,

$$a = R(1 - x^2)^{-1/3} \quad (2.6)$$

and

$$b = R(1 - x^2)^{1/6} \quad (2.7)$$

From this, we begin the proof.

Surface term

Proof. For a spherical nucleus, the surface area S_0 is

$$S_0 = 4\pi R^2$$

Substituting (2.6) and (2.7) into (2.3), we have

$$\begin{aligned} S_{\text{prolate}} &= 2\pi R(1-x^2)^{1/6} \left(R(1-x^2)^{1/6} + \frac{R(1-x^2)^{-1/3} \sin^{-1}(x)}{x} \right) \\ &= 2\pi R^2(1-x^2)^{1/6} \left((1-x^2)^{1/6} + \frac{(1-x^2)^{-1/3} \sin^{-1}(x)}{x} \right) \end{aligned}$$

Dividing by S_0 , we find

$$\frac{S_{\text{prolate}}}{S_0} = \frac{1}{2}(1-x^2)^{1/6} \left((1-x^2)^{1/6} + \frac{(1-x^2)^{-1/3} \sin^{-1}(x)}{x} \right) \quad (2.8)$$

For small x , i.e., $x \ll 1$, the following terms can be expanded as¹

$$\begin{aligned} \frac{\sin^{-1}(x)}{x} &= 1 + \frac{x^2}{6} + \frac{3x^4}{40} + \mathcal{O}(x^6) \\ (1-x^2)^{1/6} &= 1 - \frac{x^2}{6} - \frac{5x^4}{72} + \mathcal{O}(x^6) \\ (1-x^2)^{-1/3} &= (1-x^2)^{-1/3} = 1 + \frac{x^2}{3} + \frac{2x^4}{9} + \mathcal{O}(x^6) \end{aligned}$$

Therefore², (2.8) simplifies to

$$\frac{S_{\text{prolate}}}{S_0} = 1 + \frac{2}{45}x^4 + \mathcal{O}(x^6) \quad (2.9)$$

Since $\varepsilon = \frac{1}{3}x^3$, then

$$x^4 = 9\varepsilon^2$$

From this, we obtain

$$\frac{S_{\text{prolate}}}{S_0} = 1 + \frac{2}{5}\varepsilon^2 + \mathcal{O}(\varepsilon^3)$$

Therefore, the surface term in the SEMF becomes

$$\boxed{a_2 A^{2/3} \rightarrow a_2 A^{2/3} \left(1 + \frac{2}{5}\varepsilon^2 \right)} \quad (2.10)$$

to the required order, giving us (2.4). □

¹Trivial to show, taylor series.

²simplified with mathematica

Coulomb term

Proof. For a uniformly charged prolate spheroid, the Coulomb energy is [2]

$$E_C = \frac{3}{5} \frac{Q^2}{4\pi\epsilon_0 a} \frac{\tanh^{-1} x}{x} \quad (2.11)$$

For a sphere of equal volume, the corresponding Coulomb energy is

$$E_{C,0} = \frac{3}{5} \frac{Q^2}{4\pi\epsilon_0 R}$$

So, the ratio of the Coulomb energy of a prolate spheroid to that of the equal volume sphere is

$$\frac{E_{C,\text{prolate}}}{E_{C,0}} = \frac{R}{a} \frac{\tanh^{-1} x}{x}$$

Since

$$\frac{R}{a} = (1 - x^2)^{1/3},$$

then

$$\frac{E_{C,\text{prolate}}}{E_{C,0}} = (1 - x^2)^{1/3} \frac{\tanh^{-1} x}{x}. \quad (2.12)$$

For small values of x , the corresponding terms expand to the following

$$\begin{aligned} (1 - x^2)^{1/3} &= 1 - \frac{x^2}{3} - \frac{x^4}{9} + \mathcal{O}(x^6) \\ \frac{\tanh^{-1} x}{x} &= 1 + \frac{x^2}{3} + \frac{x^4}{5} + \mathcal{O}(x^6) \end{aligned}$$

Expanding the terms in (2.12) using the expansions above, we have

$$\frac{E_{C,\text{prolate}}}{E_{C,0}} = \left(1 - \frac{x^2}{3} - \frac{x^4}{9}\right) \left(1 + \frac{x^2}{3} + \frac{x^4}{5}\right) + \mathcal{O}(x^6)$$

The x^2 terms cancel, leaving

$$\frac{E_{C,\text{prolate}}}{E_{C,0}} = 1 - \frac{x^4}{45} + \mathcal{O}(x^6)$$

Again, using $x^4 = 9\varepsilon^2$, we have

$$\frac{E_{C,\text{prolate}}}{E_{C,0}} = 1 - \frac{\varepsilon^2}{5} + \mathcal{O}(\varepsilon^3)$$

Hence, for the appropriate order,

$$\boxed{a_3 \frac{Z^2}{A^{\frac{1}{3}}} \rightarrow a_3 \frac{Z^2}{A^{\frac{1}{3}}} \left(1 - \frac{1}{5} \varepsilon^2\right)} \quad (2.13)$$

which is (2.5). □

References

- [1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, [UVLé](#).
- [2] Vikram Jadhao, Zhenwei Yao, Creighton K. Thomas, and Monica Olvera de la Cruz. Coulomb energy of uniformly charged spheroidal shell systems. *Physical Review E*, 91(3):032305, 2015.

Appendices

Code

No code was used for this problem set

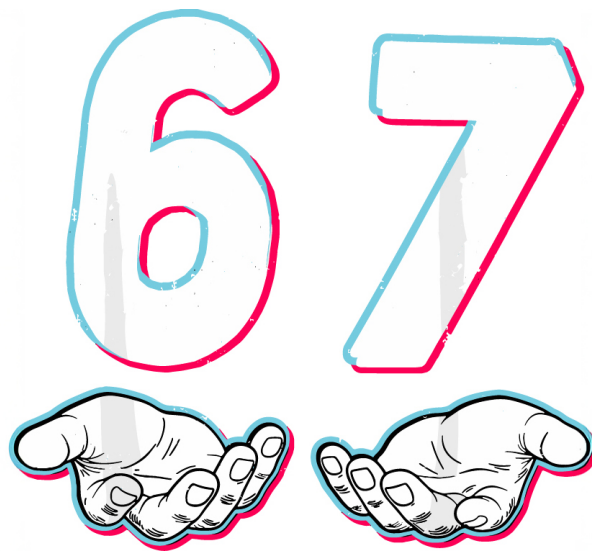


Figure 1: Problem Set 6-7... hehehe

Thank you, Mr. Kong

PHYSICA VINCIT OMNIA